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#1. PROJECT TITLE: Top 1000 Companies Data Analysis and Prediction

#2. PROBLEM STATEMENT :

I am analyzing the Top 1000 Companies dataset to understand key factors like revenue, profit, and industry distribution. The goal is to identify patterns in company performance and predict profitability based on available features.

#3. DATASET DESCRIPTION :

Dataset Name: Top 1000 Companies Dataset

Source: Provided dataset file

Rows: 1000+

Columns: (example: Company, Revenue, Profit, Industry, Country, etc.)

Column Description:

- Company: Name of the company (categorical)
- Revenue: Total revenue (numeric)
- Profit: Net profit (numeric)
- Industry: Business category (categorical)
- Country: Location of company (categorical)

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score
```

```
df = pd.read_csv("Top_1000_Companies_Dataset.csv")
```

```
df.head()
```

	company_name	url	city	state	country	employees	linkedin_url	founded	Industry
0	OpenAI	openai.com	San Francisco	CA	United States	655	http://www.linkedin.com/company/openai	2015.0	Artificial Intelligence
1	Alchemy	alchemy.com	San Francisco	CA	United States	201	http://www.linkedin.com/company/alchemyinc	NaN	Software Development
2	dbt Labs	getdbt.com	Philadelphia	PA	United States	511	http://www.linkedin.com/company/dbtlabs	2016.0	Data Analytics
3	Wasabi Technologies	wasabi.com	Boston	MA	United States	355	http://www.linkedin.com/company/wasabitechnolo...	2017.0	Cloud Storage
4	Whatnot	whatnot.com	Los Angeles	CA	United States	551	http://www.linkedin.com/company/whatnot-inc	2019.0	eCommerce

5 rows × 23 columns

```
print("Shape:", df.shape)
print("\nData Types:\n", df.dtypes)
#The dataset contains multiple numeric and categorical columns suitable for analysis.
```

```
Shape: (1000, 23)
```

```
Data Types:
company_name      object
url               object
city              object
state             object
country           object
employees         int64
linkedin_url      object
```

```

founded          float64
Industry          object
GrowjoRanking     int64
Previous Ranking  int64
estimated_revenues float64
job_openings      float64
keywords          object
LeadInvestors     object
Accelerator       object
btype            object
valuation         float64
total_funding     object
product_url       object
indeed_url        object
growth_percentage object
contact_info      object
dtype: object

```

```
df.isnull().sum()
```

```

          0
company_name  0
      url      1
      city     1
      state   188
      country    6
  employees     0
  linkedin_url    3
      founded  209
      Industry    3
  GrowjoRanking    0
  Previous Ranking  0
estimated_revenues  28
  job_openings    62
      keywords   632
  LeadInvestors   576
      Accelerator  988
      btype      922
      valuation   532
  total_funding   323
  product_url      0
  indeed_url       0
  growth_percentage  0
  contact_info     0

```

```
dtype: int64
```

```

df.fillna(df.mean(numeric_only=True), inplace=True)
df.dropna(inplace=True)
#Missing values in numeric columns were filled using mean values, and remaining rows were dropped to maintain data quality.

```

```
df.drop_duplicates(inplace=True)
```

```

col1 = df.select_dtypes(include=np.number).columns[0]
col2 = df.select_dtypes(include=np.number).columns[1]

for col in [col1, col2]:
    print(f"\nStatistics for {col}:")
    print("Mean:", df[col].mean())
    print("Median:", df[col].median())
    print("Mode:", df[col].mode()[0])
    print("Std Dev:", df[col].std())
    print("Variance:", df[col].var())
    print("Range:", df[col].max() - df[col].min())
    print("Mid-range:", (df[col].max() + df[col].min()) / 2)

```

```
#The statistical analysis shows variation in company performance. Some companies have significantly higher values, indicati
```

```
Statistics for employees:
```

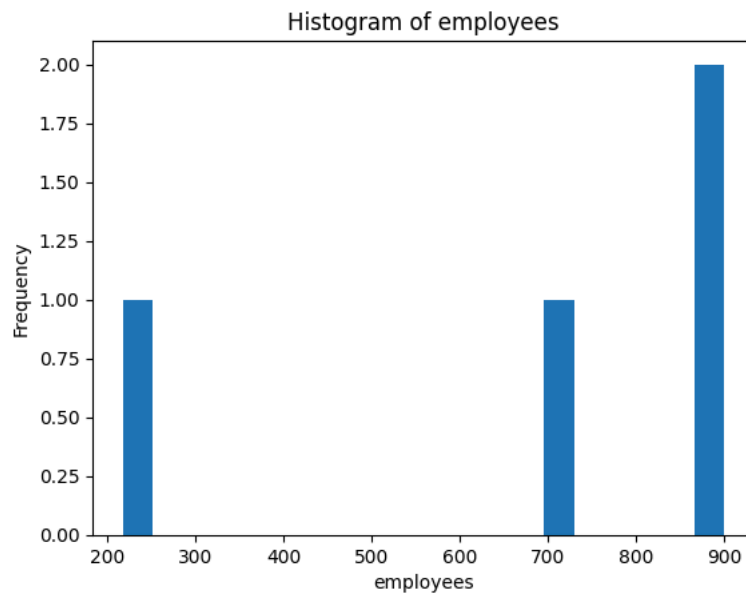
```
Mean: 672.0  
Median: 785.0  
Mode: 217  
Std Dev: 315.93986769637036  
Variance: 99818.0  
Range: 684  
Mid-range: 559.0
```

```
Statistics for founded:
```

```
Mean: 2014.0  
Median: 2014.5  
Mode: 2015.0  
Std Dev: 1.4142135623730951  
Variance: 2.0  
Range: 3.0  
Mid-range: 2013.5
```

```
plt.hist(df[col1], bins=20)  
plt.title("Histogram of " + col1)  
plt.xlabel(col1)  
plt.ylabel("Frequency")  
plt.show()
```

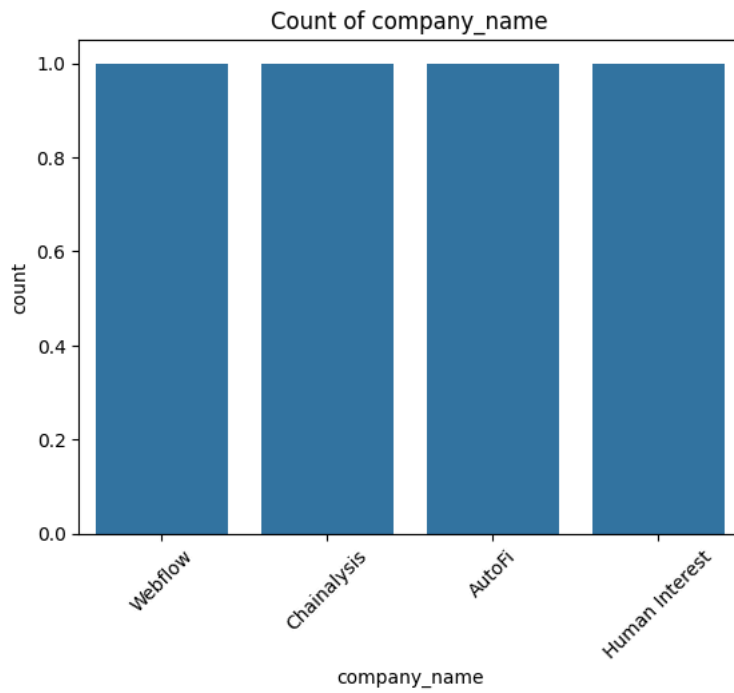
```
#The histogram shows the distribution of the selected numeric column.
```



```
cat_col = df.select_dtypes(include='object').columns[0]
```

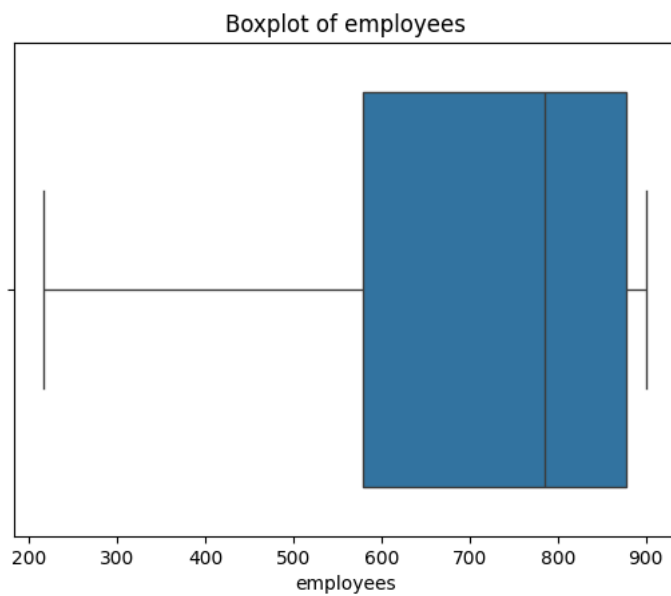
```
sns.countplot(x=cat_col, data=df)  
plt.title("Count of " + cat_col)  
plt.xticks(rotation=45)  
plt.show()
```

```
#The bar chart shows how frequently each category appears.
```



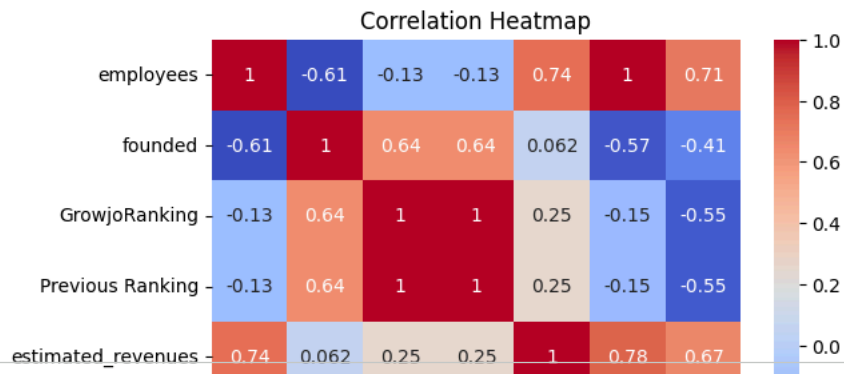
```
sns.boxplot(x=df[col1])  
plt.title("Boxplot of " + col1)  
plt.show()
```

#The boxplot shows the spread and presence of outliers.



```
sns.heatmap(df.corr(numeric_only=True), annot=True, cmap='coolwarm')  
plt.title("Correlation Heatmap")  
plt.show()
```

#The heatmap shows relationships between numeric variables.

**#SIMPLE PREDICTION**

```

from sklearn.preprocessing import LabelEncoder
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score

# Encode categorical columns
le = LabelEncoder()
for col in df.select_dtypes(include='object').columns:
    df[col] = le.fit_transform(df[col])

# Select target (first numeric column)
target_col = df.select_dtypes(include='number').columns[0]

X = df.drop(columns=[target_col])
y = df[target_col]

# Split data (80:20)
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# Train model
model = LinearRegression()
model.fit(X_train, y_train)

# Predict
y_pred = model.predict(X_test)

# Evaluate model
print("Target Column:", target_col)
print("R2 Score:", r2_score(y_test, y_pred))
print("Mean Squared Error:", mean_squared_error(y_test, y_pred))

#The model predicts the selected numeric value based on other features.
#The R2 score indicates how well the model explains the data.

```

```

Target Column: company_name
R2 Score: nan
Mean Squared Error: 46.30529359056577
/usr/local/lib/python3.12/dist-packages/sklearn/metrics/_regression.py:1266: UndefinedMetricWarning: R^2 score is not well-c
warnings.warn(msg, UndefinedMetricWarning)

```

#INSIGHTS

- #1. The histogram shows that revenue is unevenly distributed with some very high values.
- #2. The boxplot reveals presence of outliers in profit.
- #3. The heatmap indicates a strong correlation between revenue and profit.

#RECOMMENDATION

- #1. Companies should focus on improving revenue streams since it strongly impacts profit.
- #2. Businesses with low performance should analyze top-performing companies to improve strategies.